Reconstructing projective frames from their continuation graphs

Tavis Rudd

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Abstract

A projective frame in $\mathrm{PG}(2,q)$ determines a graph on the points that can be adjoined without creating three collinear points; two vertices are adjacent when they cannot be adjoined together. For $q \geq 13$, we prove that every isomorphism between these uncoloured graphs extends uniquely to a semilinear equivalence of the frames. The graph also determines q. The proof recovers tangent traces and their four centre classes, then uses compatibility between multiplication and translation to recover the field action. Completing a punctured division table makes the reconstruction algorithmic and gives polynomial-time recognition over a supplied finite field, with an independently checkable coordinate certificate. An exhaustive finite computation closes the range $5 \leq q < 13$: semilinear rigidity holds at $q = 7, 9, 11$, whereas $q = 5, 8$ each have exactly two four-pencil resolutions and an ambient subgroup of index two. The stable-range proof is independent of this computation.

Keywords. finite projective plane; arc; continuation graph; semilinear group; graph automorphism; cross ratio; $M_{0,5}$.

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1 Introduction

Let K be an arc in a finite projective plane: no line contains three points of K. A point outside K is a *legal continuation* when it can be adjoined while preserving this property. The continuation graph G_K records which pairs of legal continuations are incompatible. This graph arises naturally as the residual conflict graph of a capacity-two avoidance game, but the question studied here is static: how much of the projective geometry does the abstract graph remember?

There are three different reconstruction questions. If the ambient incidence plane is supplied, one may ask whether an ambient collineation preserving G_K must preserve K. More intrinsically, one may ask whether the abstract graph recovers the traces of tangent lines and their partition by points of K. Strongest of all is an extension question: must every abstract graph automorphism be induced by an ambient collineation? These questions are not equivalent. In particular, the fundamental theorem of projective geometry cannot be applied before enough incidence has been recovered from the graph.

Our main result settles the extension question for the first possible arc size. Every four-arc in $\mathrm{PG}(2,q)$ is a projective frame.

Theorem 1.1 (Semilinear rigidity)**.** *Let K be a projective frame in* $\mathrm{PG}(2,q)$*, where $q = p^e \geq 13$. Then, as permutation groups on the legal continuation set,*

$$\mathrm{Aut}(G_K) = \mathrm{Stab}_{\mathrm{P\Gamma L}(3,q)}(K), \qquad |\mathrm{Aut}(G_K)| = 24e.$$

Corollary 1.2 (Reconstruction and extension). *Let $K \subset \mathrm{PG}(2,q)$ and $J \subset \mathrm{PG}(2,q')$ be projective frames, where $q, q' \geq 13$. Every graph isomorphism $f : G_K \to G_J$ forces $q = q'$ and extends uniquely to a semilinear collineation carrying K to J. Thus the abstract graph determines the plane and its frame up to semilinear equivalence within this family.*

Proof. The normal form proved below gives $|V(G_K)| = (q-2)(q-3)$, which is strictly increasing for $q \geq 3$. Hence $q = q'$. Choose a projectivity h carrying K to J. The map $h^{-1}f$ is a graph automorphism, so Theorem 1.1 supplies an extension. For uniqueness, an extension acting trivially on the legal vertices fixes each recovered centre class. After frame normalization it is a field automorphism $x \mapsto x^{p^i}$. Every $x \in \mathbb{F}_q \setminus \{0,1\}$ occurs as a first coordinate, so this automorphism fixes the whole field. It is therefore the identity. □

The corollary concerns graphs promised to arise from projective frames. Section 7 also gives recognition of an arbitrary input graph against this family and checks the resulting coordinate transport. No ambient labels or colouring are supplied with the input graph.

The input is only one uncoloured binary relation. No tangent colouring, coordinate system, or ambient plane is supplied. The proof has two independent stages. First, a continuation-specific clique bound recovers the tangent traces and a second large-clique argument recovers their four centre classes. Second, after normalizing the frame, the graph becomes the agreement graph of the four maps

$$(x,y) \longmapsto x, \qquad y, \qquad x/y, \qquad (x-1)/(y-1).$$

A graph automorphism preserving the recovered classes induces permutations of the four alphabets. Compatibility with both quotient maps forces those permutations to be one field automorphism.

This coordinate model identifies the vertex set with $M_{0,5}(\mathbb{F}_q)$ and the graph with the uncoloured fibre-coincidence graph of four of the five forgetful maps to $M_{0,4}$. Bruno and Mella's theorem $\mathrm{Aut}(M_{0,n}) = S_n$ concerns biregular automorphisms of the algebraic variety [6]; it does not constrain arbitrary permutations of the finite rational point set preserving a single graph relation. The content here is precisely that the uncoloured finite graph recovers the four forgotten colours. Cross-ratio graphs supply a related automorphism and isomorphism classification [9, Definitions 3.2 and 3.4, Theorems 7.1 and 7.2]. Their vertices are the $q(q+1)$ ordered pairs of distinct projective-line points and their edges use cross-ratio orbits. Our vertex set has $(q-2)(q-3)$ elements and its edges record fibre coincidence.

The tangent-symbol representation is a linear-hypergraph reconstruction problem. Bhattacharya, Godinho, Majumder and Singhi [1, Lemmas 2.3–2.4] recover large star cliques in intersection graphs of uniform hypergraphs of bounded pair degree. For uniformity four and pair degree one, their star-clique threshold is 14; the centre-resolved argument below gives 10. Hypergraph reconstruction recovers the tangent symbols. Recovering their centre classes and forcing semilinear extension are further steps, proved here without importing a general hypergraph-recognition theorem.

The lower boundary is genuine. One-point continuation graphs are disjoint unions of cliques, every two-point continuation graph is a rook graph independent of the plane, and triangles generally admit automorphisms from the multiplicative group of the field that are not semilinear. At $q = 5$ and $q = 8$ the ambient group has index two in the graph automorphism group. The finite classification in Section 8 proves rigidity at $q = 7, 9, 11$. Thus $q \geq 13$ is the range of the uniform geometric proof, not the sharp boundary of the automorphism statement.

The paper is organized as follows. Section 2 gives the graph and its linear-hypergraph representation. Sections 3 and 4 recover tangent traces and selected centres. Section 5 records the low-cardinality obstructions. Section 6 proves Theorem 1.1. Section 9 states scope boundaries and open problems. The recognition algorithm and finite boundary are in Sections 7 and 8.

2 Continuation graphs and their intersection representation

Let $\Pi = (P, \mathcal{L})$ be a finite projective plane of order q, and let $K \subset P$ be a k-arc. Define

$$V_K = \{x \in P \setminus K : K \cup \{x\} \text{ is an arc}\}.$$

The *continuation graph* G_K has vertex set V_K, with

$$x \sim_K y \quad \Longleftrightarrow \quad \text{some line contains } x, y, \text{ and one point of } K. \tag{1}$$

Because x and y are individually legal, a line witnessing (1) contains exactly one point of K. Thus every edge has a unique *selected centre*.

For $t \in K$, let Σ_t be the $q + 2 - k$ tangent lines through t. For $x \in V_K$, put

$$S_x = \{xt : t \in K\} \subseteq \coprod_{t \in K} \Sigma_t.$$

Proposition 2.1 (Intersection representation)**.** *The sets S_x form a k-uniform linear hypergraph $\mathcal{H}_K$, and*

$$|S_x \cap S_y| = \begin{cases} 1, & x \sim_K y, \\ 0, & x \not\sim_K y. \end{cases}$$

Consequently G_K is the line graph of $\mathcal{H}_K$.

Proof. Since x is legal, xt is tangent to K for every $t \in K$, so $|S_x| = k$. Two such sets share a tangent symbol exactly when the corresponding points lie on that tangent and hence are adjacent. Distinct points cannot share two projective lines, proving linearity. □

After choosing a label for each pencil Σ_t, the map

$$x \longmapsto (xt)_{t \in K} \in \prod_{t \in K} \Sigma_t$$

is an injective nonlinear code of length k. Distinct words agree in at most one coordinate, and adjacency means agreement in exactly one. No linearity or additivity is asserted, so standard MacWilliams extension theorems do not apply; compare [8].

3 Intrinsic recovery of tangent traces

For a tangent line ℓ, write $B_\ell = \ell \cap V_K$. This is a clique of G_K, but a clique need not a priori lie on one tangent: different edges may have different selected centres.

Lemma 3.1 (Non-tangent clique bound)**.** *Assume $k \geq 2$. If a clique C of G_K is not contained in a tangent trace, then*

$$|C| \leq k(k - 2) + 1.$$

Proof. Fix $x \in C$ and partition $C \setminus \{x\}$ into sets C_t, where $y \in C_t$ when the edge xy is centred at $t \in K$. For a nonempty C_t, choose $z \in C$ outside the line xt; such a z exists because C is not contained in that trace. Let u be the centre of xz.

For $y \in C_t$, let s_y be the centre of zy. The centres s_y are distinct: otherwise two distinct points of xt would also lie on one line through z. Moreover $s_y \notin \{t, u\}$, since either equality forces z or y onto two distinct lines through x. Hence $|C_t| \leq k-2$. There are at most k parts, and therefore $|C| - 1 = \sum_t |C_t| \leq k(k-2)$. ☐

Deza's weak-Δ-system theorem gives the related generic bound $k^2 - k + 1$ for a non-sunflower family of k-sets [7]. The improved constant above uses the resolution of every continuation edge by a selected centre.

Theorem 3.2 (Tangent-trace recovery)**.** *If*

$$q - \binom{k-1}{2} > k(k-2) + 1, \tag{2}$$

then the tangent traces are exactly the maximal cliques of G_K with more than $k(k-2) + 1$ vertices. Hence every abstract graph isomorphism $G_K \to G_J$ canonically induces an isomorphism of the uncoloured tangent-trace hypergraphs.

Proof. A tangent contains q points besides its centre. At most $\binom{k-1}{2}$ of them lie on secants determined by the other selected points, so every tangent trace has more than the threshold in the statement. Lemma 3.1 says that every clique above that threshold lies in one tangent trace. Such a trace is maximal: a vertex adjacent to its whole trace would create a larger clique and hence would itself lie on the same tangent. Conversely every maximal clique above the threshold is therefore the full trace containing it. ☐

4 Intrinsic recovery of selected centres

Assume Theorem 3.2 applies. Form a graph D_K whose vertices are the recovered tangent traces, joining two traces when they are disjoint as subsets of V_K.

Theorem 4.1 (Centre-class recovery)**.** *Put*

$$T = q + 2 - k, \qquad c = \binom{k-2}{2}.$$

If $T > kc$, then the k classes of tangent traces having a common selected centre are exactly the maximal cliques of D_K with more than kc members. Thus G_K intrinsically recovers its tangent traces, their partition into selected centres, and every tangent-incidence triple (t, x, y) with $x, y \in V_K$.

Proof. The T traces centred at one $t \in K$ are pairwise disjoint. Fix a trace A centred at t and another centre $u \neq t$. A tangent trace at u can be disjoint from A only when the two ambient tangent lines meet at an illegal point. That point cannot lie on a secant involving t or u; it must lie on one of the c secants whose endpoints are both in $K \setminus \{t, u\}$. Each such secant determines at most one tangent through u, so at most c traces centred at u are disjoint from A.

Now let a clique of D_K use at least two centre classes, and choose traces A, B from distinct classes. The trace A bounds the contribution from each other class by c, while B gives the same bound for the class of A. The mixed clique therefore has at most kc members. Under $T > kc$, every larger clique lies in one centre class, and maximality recovers the whole class. ☐

For reference, the resulting sufficient integer thresholds are

k	3	4	5	6	8
q	7	13	23	41	127.

They are method bounds and are not expected to be sharp. The disjointness argument is plane-specific: in higher dimension, tangent lines with different centres may be skew.

5 Why extension cannot begin before four points

Proposition 5.1 (One point)**.** *For a one-point arc in* $\mathrm{PG}(n, q)$*, the continuation graph is a disjoint union of* $1 + q + \cdots + q^{n-1}$ *copies of* K_q*. This structure permits permutations of individual clique fibres; semilinear rigidity does not hold uniformly for one-point arcs.*

Theorem 5.2 (Two points)**.** *If* $K = \{a, b\}$ *in any projective plane of order* q*, then*

$$G_K \cong K_q \,\square\, K_q.$$

In particular, its isomorphism type is independent of the projective plane.

Proof. The legal points are precisely the q^2 points outside ab. Map such a point x to the ordered pair of tangent lines (ax, bx). Every choice of one tangent at a and one at b meets in one legal point, and two points are adjacent exactly when they share one of these tangent lines. This is rook adjacency. □

Theorem 5.3 (Three points)**.** *Let* K *be a triangle in* $\mathrm{PG}(2, q)$*. With its vertices normalized to the coordinate points, the legal vertices are* $(x : y : 1)$ *with* $x, y \in \mathbb{F}_q^*$*, and*

$$(x, y) \sim (x', y') \iff x = x' \text{ or } y = y' \text{ or } x/y = x'/y'. \tag{3}$$

Every automorphism α *of the multiplicative group* $\mathbb{F}_q^*$ *therefore induces the graph automorphism* $(x, y) \mapsto (\alpha(x), \alpha(y))$*. This is not induced by a semilinear collineation unless* α *is a field automorphism.*

Proof. The three equalities in (3) are the three tangent-pencil parameters. A multiplicative-group automorphism preserves equality and quotient. If an ambient semilinear collineation induces this map, preservation of the three pencil partitions forces it to fix the triangle vertices. In normalized coordinates it has the form $(x, y) \mapsto (A\sigma(x), B\sigma(y))$ for a field automorphism σ. Since $\alpha(1) = 1$, equality with the displayed graph automorphism forces $A = B = 1$ and $\alpha = \sigma$. □

For example, inversion on $\mathbb{F}_5^*$ is nonambient. These examples separate intrinsic centre recovery from ambient semilinear extension.

6 The four-point frame

Normalize the frame as

$$K = \{(1 : 0 : 0), (0 : 1 : 0), (0 : 0 : 1), (1 : 1 : 1)\}.$$

Proposition 6.1 (Four-coordinate normal form). *The legal continuation points are*

$$\Omega = \{(x, y) : x, y \notin \{0, 1\},\ x \neq y\}.$$

The map

$$(x, y) \longmapsto \left(x, y, \frac{x}{y}, \frac{x-1}{y-1}\right) \tag{4}$$

is injective, and two vertices are adjacent exactly when their images agree in one coordinate.

Proof. The six frame secants have equations $X = 0$, $Y = 0$, $Z = 0$, $X = Y$, $X = Z$, and $Y = Z$. After normalizing $Z = 1$, avoiding them gives the displayed set Ω. Direct equations for lines through the four frame points give, up to reparametrization, the four pencil parameters in (4). $\square$

Write $D = \mathbb{F}_q \setminus \{0, 1\}$. For $q \geq 13$, Theorems 3.2 and 4.1 recover the four coordinate partitions. After composing with the unique projectivity inducing the corresponding permutation of the frame, a graph automorphism preserves all four partitions. Its alphabet permutations satisfy

$$\frac{\alpha(x)}{\beta(y)} = \gamma(x/y), \tag{5}$$

$$\frac{\alpha(x)-1}{\beta(y)-1} = \delta\left(\frac{x-1}{y-1}\right) \tag{6}$$

for all distinct $x, y \in D$.

Lemma 6.2 (Punctured multiplicative isotopy). *Suppose permutations α, β, γ of D satisfy* (5) *for all distinct $x, y \in D$, with $q \geq 5$. Then $\alpha = \beta = \gamma = \chi$, where χ extends to an automorphism of the multiplicative group $\mathbb{F}_q^*$.*

Proof. The map $(x, y) \mapsto (\alpha(x), \beta(y))$ sends Ω injectively to itself and hence bijectively to itself. The row with first coordinate x omits column x; its image omits $\beta(x)$, whereas the complete output row with first coordinate $\alpha(x)$ omits $\alpha(x)$. Thus $\alpha = \beta =: h$.

Writing $x = ry$, equation (5) becomes

$$h(ry) = \gamma(r)h(y) \tag{7}$$

whenever $r, y, ry \neq 1$. Extend $\gamma(1) = 1$. If $r, s, rs \neq 1$, choose $y \notin \{1, s^{-1}, (rs)^{-1}\}$; this is possible since $|\mathbb{F}_q^*| \geq 4$. Applying (7) twice gives $\gamma(rs) = \gamma(r)\gamma(s)$. The inverse case follows by choosing $y \notin \{1, r\}$ and comparing $h(y)$ with $h(rr^{-1}y)$. Hence γ extends multiplicatively.

Equation (7) also makes $h(x)/\gamma(x)$ constant on D, say $h(x) = A\gamma(x)$. Since both h and γ permute D, one has $AD = D$. But $AD = \mathbb{F}_q^* \setminus \{A\}$ and $D = \mathbb{F}_q^* \setminus \{1\}$, so $A = 1$. $\square$

Lemma 6.3 (Shifted isotopy forces Frobenius). *Assume $q \geq 5$. If permutations $\alpha, \beta, \gamma, \delta$ of D satisfy both* (5) *and* (6)*, then all four are restrictions of one field automorphism of $\mathbb{F}_q$.*

Proof. Lemma 6.2 gives $\alpha = \beta = \gamma = \chi$ for a multiplicative-group automorphism χ. Define $\eta(z) = 1 - \chi(1 - z)$ on D. Substitution of $x = 1 - z$ and $y = 1 - w$ in (6) yields $\eta(z)/\eta(w) = \delta(z/w)$. A second application of Lemma 6.2 gives $\eta = \delta =: \lambda$, where λ is another multiplicative-group automorphism.

Write $\chi(x) = x^m$ and $\lambda(x) = x^n$, with $1 \leq m, n \leq q - 2$ and both exponents coprime to $q - 1$. The definition of η says

$$(1 - z)^m + z^n = 1$$

for every $z \in D$, and the identity also holds for $z = 0, 1$. The polynomial $(1-X)^m + X^n - 1$ has all q field elements as roots but degree at most $q-2$, so it vanishes identically. Comparing degrees gives $m = n$, and then $(1-X)^m = 1 - X^m$. All intermediate binomial coefficients vanish in characteristic p. Write $m = p^r u$ with $p \nmid u$. If $u > 1$, the coefficient of X^{p^r} in $(1-X)^m = (1-X^{p^r})^u$ is $-u \neq 0$, contradicting the vanishing of intermediate coefficients. Thus $u = 1$. Therefore all four maps are the same Frobenius automorphism. □

Proof of Theorem 1.1. For $k = 4$, the hypotheses of Theorems 3.2 and 4.1 hold when $q \geq 13$. Thus every graph automorphism permutes the four recovered centre classes. Every permutation of a projective frame is realized by a unique projectivity. Compose by its inverse to obtain an automorphism preserving all four classes. In the normal form of Proposition 6.1, its symbol permutations satisfy (5) and (6); Lemma 6.3 makes them one field automorphism. This is induced by a semilinear collineation fixing the normalized frame.

Conversely, every semilinear frame stabilizer preserves legal points and collinearity, so it acts on G_K. The projective stabilizer of a frame is S_4, and the field automorphism group has order $e = [\mathbb{F}_q : \mathbb{F}_p]$. □

Remark 6.4 (The small-order boundary). The isotopy lemmas already apply for $q \geq 5$; the bound $q \geq 13$ enters only when the uncoloured graph is required to recover its four coordinate partitions. At $q = 5$, $G_K \cong K_6 \setminus 3K_2$, whose automorphism group has order 48, twice the projective frame stabilizer. Section 8 settles the remaining orders $7, 8, 9, 11$ by exact finite enumeration.

7 Recognition and coordinate certificates

The proof can be made constructive without searching through all permutations of the field alphabet. First recover the pencils. Then the missing entries in three pencils complete a group division table, leaving only the possible images of a cyclic generator to test.

Lemma 7.1 (Four-point trace seeds). *Let K be a four-point frame and B a tangent trace. A vertex outside B has at most three neighbours in B. Consequently, if Q consists of four vertices of B, then*

$$B = Q \cup \bigcap_{v \in Q} N(v).$$

For $q \geq 13$, the sets on the right that are cliques of size $q-3$, as Q ranges over four-cliques, are exactly the tangent traces.

Proof. Write t for the centre of B. For an outside vertex z, a neighbour on B must lie on one of the three lines zu with $u \in K \setminus \{t\}$. Each such line meets the line of B at most once. Thus no outside vertex is adjacent to four vertices of B. Conversely all other vertices of B are adjacent to those four vertices. Every trace has size $q-3$ by the normal form. Any resulting clique of that size has at least ten vertices, so Lemma 3.1 places it in a tangent trace; equal sizes give equality. □

Theorem 7.2 (Polynomial recognition over a supplied field). *Given a graph G with n vertices and a model of $\mathbb{F}_q$, $q \geq 13$, one can decide in polynomial time whether G is a frame continuation graph and, if it is, output a bijection*

$$\phi : V(G) \longrightarrow \{(x, y) \in (\mathbb{F}_q \setminus \{0, 1\})^2 : x \neq y\}$$

that preserves and reflects adjacency. A coarse bound for the procedure below is $O(n^6)$ elementary adjacency and field operations after rejecting $n \neq (q-2)(q-3)$. A supplied coordinate certificate can be checked using $O(n^2)$ field operations and adjacency tests.

Proof. Enumerate four-cliques and apply Lemma 7.1, retaining only clique closures of size $q-3$. Reject unless there are $4(q-2)$ traces. Construct their disjointness graph. Recover centre classes as the large maximal cliques from Theorem 4.1; concretely, enumerate five-cliques and take their closed common neighbourhoods. In a true frame graph every such five-clique is contained in one centre class, and a trace outside that class is disjoint from at most one member of it, so this closure is exactly the class. Reject unless there are four classes of $q-2$ traces partitioning the recovered traces and each class partitions $V(G)$. These tests do not alone certify recognition.

Order the four classes. The first two provide rows and columns. In a true frame graph each row meets each column in one vertex, except for one missing column; missing columns pair the row and column alphabets. Call their common alphabet D. Each row also omits exactly one symbol of the third class. Identify that missing symbol with its row label. In the normal form this is the identity $x/1 = x$: row x omits the quotient value x.

The third symbol at cell (x, y) now gives x/y for distinct $x, y \in D$. Adjoin a symbol 1, put $x/x = 1$ and $x/1 = x$, and define $1/y$ as the unique symbol of D missing from column y. Set $1/1 = 1$. Reject inconsistent or nonunique entries. For a frame this completes the full division table of $\mathbb{F}_q^*$.

Set $ab = a/(1/b)$. Try each element as a cyclic generator, checking its powers exhaust the $q-1$ symbols and that the division table agrees with subtraction of exponents modulo $q-1$. This recognizes a cyclic group directly, without assuming associativity of an unchecked table. Map a generator to each primitive element of the supplied field in turn. For each map, form (x, y) coordinates and check that the fourth class is exactly the fibre partition of $(x-1)/(y-1)$. Finally check that the coordinates are a bijection onto the displayed set and that adjacency agrees for every pair. Output a passing map, or reject if none passes.

For a true frame any ordering of the four pencils is realized by a projectivity, so the completed table and at least one generator image pass these tests. Conversely the final check itself proves that a passing map is an isomorphism; no promised geometry is needed for soundness.

There are at most $O(n^4)$ seeds, with $O(n^2)$ work per closure and clique check. After the trace-count check the disjointness graph has $O(\sqrt{n})$ vertices, so its five-seed enumeration and checks cost $O(n^{7/2})$. Table completion, cyclic-table checks and all generator images take polynomial time bounded by $O(q^3 + qn^2)$ here. The initial vertex count gives $q = O(\sqrt{n})$, establishing the stated coarse bound. The last pairwise adjacency check is also the independent certificate verification procedure. □

Proposition 7.3 (Size of the coordinate representation)**.** *A frame graph has* $(q-2)(q-3)$ *vertices, degree* $4(q-4)$*, and* $2(q-2)(q-3)(q-4)$ *edges. Four field symbols per vertex determine adjacency by equality in one coordinate.*

Proof. Each coordinate fibre in the normal form has $q-3$ vertices. Distinct fibres through a vertex meet only at that vertex, so its neighbours split into four sets of size $q-4$. Count edges by summing degrees. □

The coordinate representation therefore uses $\Theta(q^2)$ field symbols instead of $\Theta(q^3)$ explicit edges. This is a storage statement, not a speedup claim for every graph operation. In particular, enumerating all edges still has its output cost. The supplied reference implementation and timing protocol are described in the verification appendix.

8 The finite boundary

A *four-pencil resolution* is an unordered set of four partitions of $V(G)$, each into $q-2$ cliques of size $q-3$, such that every edge belongs to exactly one block across the four partitions. Blocks need not be maximal cliques. The geometric pencils give one such resolution.

Proposition 8.1 (Exact small-order census)**.** *For the indicated frame graphs the following table is exact. Here b counts all $(q-3)$-cliques, p counts their partitions of the vertex set, and r counts four-pencil resolutions.*

q	$\lvert V\rvert$	$\lvert\operatorname{Aut}(G_K)\rvert$	b	p	r
5	6	48	12	8	2
7	20	24	180	1352	1
8	30	144	48	16	2
9	42	48	28	4	1
11	72	24	36	4	1

At $q = 5, 8$ the two resolutions form one orbit, and the stabilizer of the geometric resolution is exactly the ambient semilinear group, of index two. At $q = 7, 9, 11$ every graph automorphism is ambient. The two resolutions share all twelve blocks at $q = 5$ and no blocks at $q = 8$.

Computational proof and completeness. The certificate bundle named in Appendix A enumerates all cliques of size $q-3$, all exact covers of the vertex set by such cliques, and all exact covers of the edge set by the resulting partitions. At each recursive step an uncovered vertex or edge is chosen and every available covering block is explored. This enumerates each unordered cover once and terminates because the uncovered set strictly decreases. A second implementation enumerates fixed-size cliques directly instead of deriving them from maximal cliques. The two complete enumerations agree by canonical digests.

Graph automorphism generators are computed with Sage's graph algorithm. Their generated groups are explicitly enumerated and every permutation is checked against the edge set. An independent nauty computation gives the same full group orders, establishing the upper bound as well as the witnessed subgroup. The action on all resolutions is checked explicitly. Finally, the isotopy lemmas apply for every $q \geq 5$: an automorphism stabilizing the geometric resolution, after a frame permutation, is Frobenius. Hence its stabilizer is precisely the ambient group. The bundle supplies an explicit representative of the other coset in each exceptional case; multiplying it by the ambient group describes every nonambient automorphism. □

Together with Theorem 1.1, the census proves semilinear rigidity for every prime power $q \geq 5$ except $q = 5, 8$. This combined statement has a finite computational premise at $q = 7, 9, 11$; the theorem for $q \geq 13$ does not. The census concerns *four-point frames.* A six-point Clebsch seed over $\mathbb{F}_{11}$ is a different reconstruction problem.

At $q = 7$ there are 1352 individual clique partitions but only one compatible four-pencil resolution. Thus uniqueness of the full resolution can hold even when its individual candidate partitions are plentiful. At $q = 8$ the two-resolution orbit is exactly the obstruction to extending all graph automorphisms semilinearly.

9 Scope, a stronger complex, and open problems

The theorem reconstructs the four tangent-pencil classes and proves that all graph automorphisms are ambient. It does not say that the binary graph alone reconstructs every line of the projective plane for a general arc, and it says nothing about the outcome or Sprague–Grundy value of an avoidance game.

There is a stronger residual object,

$$\Delta_K = \{X \subseteq V_K : K \cup X \text{ is an arc}\}.$$

Its minimal nonfaces are the incompatible pairs whose joining line contains one selected point, together with the collinear triples whose line contains no selected point. Thus Δ_K retains line-incidence information that G_K discards. Under explicit large-order hypotheses, this information can be used to complete the complement of the secant arrangement and then recover K from the multiplicities of the restored secants. This belongs to the classical neighbourhood of projective-plane complement and pseudo-complement embedding theorems [2, 5, 3, 4]; it is not a contribution claimed in this paper.

The present arguments isolate two extremal parameters. Let $m(k)$ be the largest clique in a k-arc continuation graph that is not contained in one tangent trace, and let $r(k)$ be the largest disjointness clique of tangent traces using at least two selected centres. We proved

$$m(k) \le k(k-2) + 1, \qquad r(k) \le k\binom{k-2}{2}.$$

Problem 9.1. Determine $m(k)$ or $r(k)$ exactly for an infinite family of k, or improve their orders of growth. Any improvement immediately lowers the intrinsic recovery threshold.

Problem 9.2. Give a computation-free proof of the boundary in Proposition 8.1, especially the two-resolution phenomenon at $q = 8$ and uniqueness despite the many individual partitions at $q = 7$.

Problem 9.3. For larger arcs, determine natural hypotheses under which recovery of the tangent traces and their centre classes forces every graph isomorphism to extend semilinearly.

A Verification and reproducibility

The stable-range results have written proofs and use no finite census. No theorem in this draft is claimed to be Lean-formalized. The small-order census is an exact computer-assisted result, independently replayed using different graph and clique algorithms; its trust base includes Python, SageMath, nauty, and their arithmetic and graph implementations.

The public bundle contains `verification/boundary.json`, its SHA-256 manifest, the generator, the independent replay, and the coordinate-certificate checker. From the repository root, `make check` checks the statement/evidence links and replays the finite census and deterministic recognition tests. `make boundary` regenerates the census with SageMath; `make benchmark` records a fresh timing sample. The verification README specifies the field models, versions, exact domains, relabelling seeds and trusted boundaries.

The reference recognizer supplies prime-field models and explicit models of orders $8, 9$; stable-range transport tests use $q = 13, 17, 19$. The mathematical recognition theorem allows any supplied finite field. Checks include relabelled inputs, invalid coordinate certificates, deleted edges, and degree-preserving edge switches. Recorded timings compare reconstruction with generic graph isomorphism on the same relabelled inputs, excluding graph construction. They are single-run measurements, not a claim that reconstruction is faster.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author's direction, the systems assisted throughout with proof exploration, literature research, symbolic and finite computation, code and formal-proof development, verification, and manuscript drafting and revision. The author assumes responsibility for the manuscript and its accompanying artifacts; this first draft remains subject to author and external review.

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